

# HEBA SLEIMAN

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## Education

**M2 MoSIG — Master of Science in Informatics, Artificial Intelligence specialization**

2026 – 2027 (expected)

Université Grenoble Alpes / Grenoble INP, Grenoble, France

**B.S. Computer Science, Minor in Data Analytics — Graduated with Honors**

May 2025

Lebanese American University, Beirut, Lebanon

## Experience

**Software Engineering Intern — simBrAI Solutions**

Jul 2024

Beirut, Lebanon

- Performed data preprocessing and feature engineering, then developed and integrated ML models into production web applications
- Built full-stack features — email routing, real-time notifications, interactive UI — using ASP.NET MVC and Bootstrap

## Projects

**Real-Time Ergonomic Risk Assessment in VR (RULA)** — Master's Research Internship, G-SCOP Lab, Université Grenoble Alpes, supervised by Dr. Mathieu Delangle (Python, scikit-learn, Unreal Engine 5, OptiTrack, TCP sockets)

- Built an end-to-end closed-loop system: OptiTrack + Meta Quest motion capture → Unreal Engine joint-angle computation → labeled dataset → classifier served over a Python TCP server → real-time risk visualization and workstation adaptation
- Trained and compared Random Forest and SVM classifiers on 6 joint angles per side, selecting Random Forest for its stronger recall on high-risk postures, and built a proof-of-concept loop where the workstation's table height adapts automatically once a tracked joint enters a high-risk angle range

**Resume-to-JD Matching Agent** (Python, Anthropic Claude API, Pydantic, Streamlit)

- Built an LLM pipeline using Claude tool-use / function calling to extract structured candidate profiles from resume PDFs and score them against a job description
- Implemented a bias-audit module that re-scores candidates with demographic signals swapped and flags score drift, plus an eval harness for probabilistic LLM output
- Deployed the agent publicly on Render with a Streamlit interface

**Tasklet Support Agent** (Python, Anthropic Claude API, SQLite)

- Built a multi-turn conversational support agent using Claude tool-use, with per-user data scoping enforced at the service layer rather than by the LLM
- Ran adversarial refusal red-teaming (jailbreak attempts, fabricated authority, injected fake instructions) to validate the agent's refusal behavior held under pressure

**COMPAS Fairness Analysis** — (Python, pandas, scikit-learn)

- Analyzed racial and gender bias in the ProPublica COMPAS recidivism dataset, computing fairness metrics (PPV, FPR, FNR) and calibration curves across demographic groups to diagnose the trade-off between calibration and equalized-odds fairness
- Trained and benchmarked four classifiers (Logistic Regression, Decision Tree, Random Forest, Gradient Boosting) against the COMPAS score on accuracy and AUC-ROC, and evaluated whether the same racial fairness gaps persisted in self-trained models

## Publications

**Numerical Solutions of the Nonlinear Poliovirus Model: A Radial Basis Dual Layered Neural Network Approach**

Biomedical Signal Processing and Control, 2025 — with Dr. Zulqurnain Sabir, M.A. Abdelkawy & Fatima Hassouna

- Co-developed a dual-layer radial basis function neural network (18 and 34 neurons, trained via Bayesian regularization) to numerically solve a nonlinear four-compartment (susceptible-infected-exposed-vaccinated) poliovirus transmission model
- Validated solver accuracy against a Runge-Kutta-generated dataset using error histograms, regression coefficients, and training-performance diagnostics

## Technical Skills

- **Programming:** Python, Java, C, SQL, HTML, CSS, PHP
- **Machine Learning:** scikit-learn, Random Forest, SVM, Logistic Regression, Gradient Boosting, fairness metrics (PPV, FPR, FNR), calibration analysis, AUC-ROC
- **Generative AI & LLMs:** Anthropic Claude API, LLM tool use / function calling, prompt engineering, structured-output validation (Pydantic), LLM evals & red-teaming
- **Simulation & Real-Time Systems:** Unreal Engine 5, OptiTrack motion capture, TCP sockets
- **Data & Visualization:** pandas, Tableau, SQLite, Oracle SQL, data modeling
- **Tools & Deployment:** Git/GitHub, Streamlit, Render, ASP.NET MVC
- **Languages:** Arabic (native), English (fluent)